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Studierichting: Informatica
Oefeningenreeks 3F nr. 1.12

$$f(x) = \frac{x^3 + 4x^2 + x + 1}{x - 3}$$

Verticale Asymptoot:

$x - 3 = 0 \Leftrightarrow x = 3$

$$\lim_{x \to 3+} \frac{x^3 + 4x^2 + x + 1}{x - 3} = +\infty$$

$$\lim_{x \to 3-} \frac{x^3 + 4x^2 + x + 1}{x - 3} = -\infty$$

Verticale asymptoot op $x = 3$

x		3	
$f(x)$	+	|	−

Horizontale Asymptoot:

$$\lim_{x \to \infty} \frac{x^3 + 4x^2 + x + 1}{x - 3} = \lim_{x \to \infty} \frac{x^3}{x} = \lim_{x \to \infty} x^2 = +\infty$$

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Schuine Asymptoot:

$$\lim_{x \to \infty} \frac{\frac{x^3 + 4x^2 + x + 1}{x - 3}}{x} = \lim_{x \to \infty} \frac{x^3 + 4x^2 + x + 1}{x^2 - 3x} = \lim_{x \to \infty} \frac{x^3}{x^2} = \lim_{x \to \infty} x = \infty$$

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References